Please write clearly in block capitals.

Centre number | | | | | Candidate number | | | | |

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Functional Skills Level 2
MATHEMATICS
Section A: Non-Calculator Paper
Section B: Calculator Paper

Paper A/B

Time allowed: 30 minutes / 1 hour 30 minutes

Materials

For this paper you must have:

- mathematical instruments.

You **must not** use a calculator for section A.
You may use a calculator for section B.

Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer all questions.
- You must answer the questions in th x around each page or on blank pages.
- Do all rough work in this book. Cross ed.
- State the units of your answer where

Information

- The marks for questions are shown in
- The maximum mark for section A is 15
- The maximum mark for this paper is 6
- You may ask for more answer paper, gr
  These must be tagged securely to this

Advice

- In all calculations, show clearly how you work out your answer.

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60 / 60 Marks (100% Pass)

· Assessment: Practice Paper (60 Marks)
· Complete step-by-step worked models
· Official Mark Scheme Allocations [M1 / A1]
· Exam Board: Open Awards (Ofqual Regulated)

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SECTION A : Non-Calculator

Answer ALL questions.
Write your answers in the spaces provided.

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1 Write the following numbers in order of size.

3.2 3.27 3.72 3.702 3.02

[1 mark]

Compare place values (tenths, hundredths, thousandths):

3.02 < 3.2 < 3.27 < 3.702 < 3.72 **[1 mark]**

Answer **3.02, 3.2, 3.27, 3.702, 3.72** ✓

2 Work out

$$1\frac{3}{5} + 2\frac{1}{4}$$

[2 marks]

Convert mixed numbers to improper fractions with common denominator (20):

1 3/5 = 8/5 = 32/20 and 2 1/4 = 9/4 = 45/20 **[M1]**

32/20 + 45/20 = 77/20 = 3 17/20 **[A1]**

Answer **77/20 (or 3 17/20)** ✓

3 The distance between two towns on a map measures 8 cm. The map has a scale of 1: 25000

What is the actual distance between the two towns in km?

[1 mark]

Multiply map distance by scale, then convert cm to m to km:

8 cm x 25 000 = 200 000 cm --> 2 000 m --> 2 km **[1 mark]**

Answer **2 km** ✓

Turn over for next question

4

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4 Calculate $6.402 + 3.545$

[1 mark]

Align decimals vertically by place value:

6.402 + 3.545 = 9.947 [1 mark]

Answer **9.947** ✓

5 The distance an object has moved from its starting point, d, after an amount of time, t, is given by the formula

$$d = 29t^2$$

where d is measured in metres and t is measured in seconds.

Estimate the distance the object has travelled after 5.1 seconds.

[3 marks]

Round values to 1 significant figure: 29 ≈ 30, 5.1 ≈ 5 [M1]

Substitute into formula: d ≈ 30 × (5)² = 30 × 25 [M1]

30 × 25 = 750 metres [A1]

Answer **750 m** ✓

6 Calculate $4335 \div 6$

[1 mark]

Short division: 4335 ÷ 6 = 722 remainder 3 = 722.5

6 | 4 3¹ 3¹ 5 . ³0 = 0 7 2 2 . 5 [1 mark]

Answer **722.5** ✓

Turn over for next question

5

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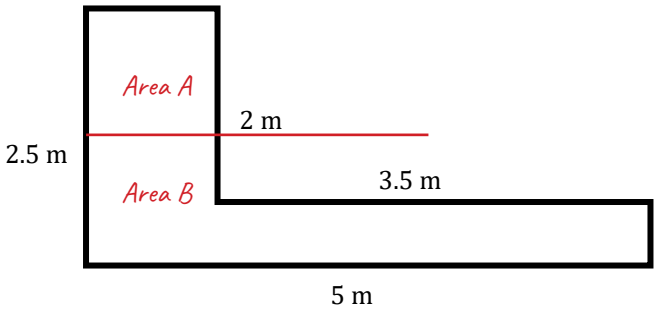
7 Some painters are planning on painting a shape on a wall. They have drawn a sketch of the painting below.



Calculate the area the painters are planning on painting.

[3 marks]

Split compound shape horizontally into two simple rectangles:

- **Top rectangle (A): 2 m × (2.5 m − 1 m) = 2 m × 1.5 m = 3.0 m²** [M1]
- **Bottom rectangle (B): 5 m × (2.5 m − 2 m) = 5 m × 0.5 m = 2.5 m²**

Total Area = 3.0 m² + 2.5 m² = 5.5 m² [M1 A1]

Answer **5.5 m²** ✓

8 It would take 7 hours for 3 people to paint the shape on the wall. How long would it take to paint the shape on the wall if another person joins, assuming that they all paint at the same rate?

[3 marks]

Inverse proportion: Calculate total person-hours required:

Total work = 7 hours × 3 people = 21 person-hours [M1]

With 1 more person joining (3 + 1 = 4 people):

Time = 21 person-hours ÷ 4 people = 5.25 hours (5h 15m) [M1 A1]

Answer **5.25 hours (or 5 hours 15 mins)** ✓

End of Section A

6

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SECTION B : Calculator

Answer ALL questions.
Write your answers in the spaces provided.

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9 Write in digits five hundred and seven thousand, seven hundred and twenty three.

[1 mark]

Five hundred and seven thousand (507 000) + seven hundred and twenty three (723):

Answer **507,723** ✓ *[1 mark]*

10 A manufacturing company produce metal fixings for a DIY retailer.
On one day the manufacturer produces 23040 fixings, of which 1920 are faulty.

What fraction of the fixings produced on this day were defective?
Give your answer in its simplest form.

[2 marks]

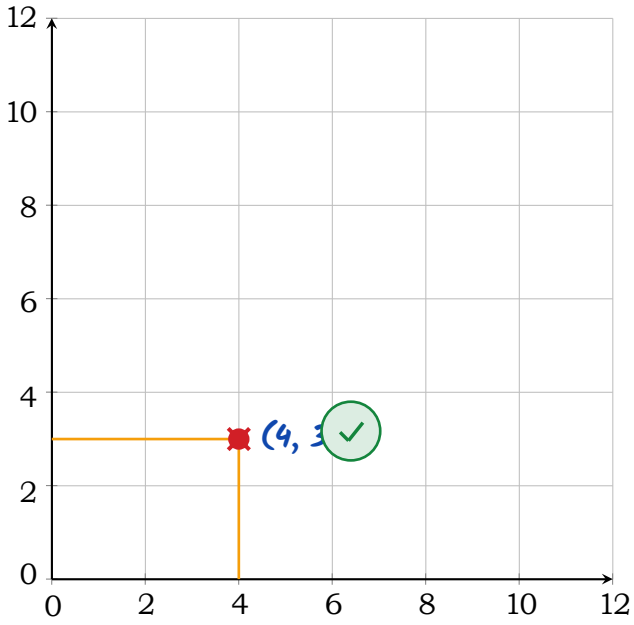
Express as a fraction and simplify fully by dividing by 1920:

Fraction defective = 1920 / 23040 *[M1]*

= 192 / 2304 = 1 / 12 *[A1]*

Answer **1 / 12** ✓

11 Plot the coordinate (4,3) on the graph below.



[1 mark]

[1 mark]

Turn over for next question

4

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12 Darren has been given £6000 from his parents to put towards a car in the future. He is looking for the best savings account to put his money in for 2 years.

He sees these two savings accounts:

YTwo savings account	**Standard savings account**
Save for 2 years and receive 4.25% interest	2.5% interest added at the end of every year

He says that he would have over £50 more after 2 years if he used the Standard savings account instead of the YTwo savings account.

Is he correct?

Show your working.

[5 marks]

Step 1: Calculate return from YTwo Savings (4.25% fixed over 2 years):

- **Total = £6000 × 1.0425 = £6255.00 (Interest = £255.00)** **[M1]**

Step 2: Calculate return from Standard Savings (2.5% compound each year):

- **Year 1: £6000 × 1.025 = £6150.00** **[M1]**
- **Year 2: £6150 × 1.025 = £6303.75 (Interest = £303.75)** **[M1]**

Step 3: Difference between the two savings accounts:

- **Difference = £6303.75 − £6255.00 = £48.75** **[M1 A1]**

Step 4: Check Darren's statement ('over £50 more'):

- **£48.75 < £50.00, so Darren is incorrect (£1.25 short).** **[A1]**

Answer **No, Darren is incorrect. Standard gives £48.75 mo… £50).**

Turn over for next question

5

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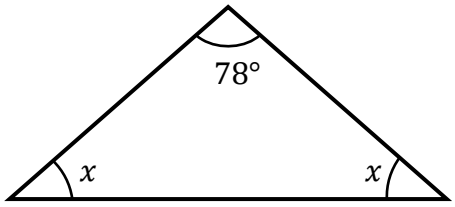
13 The diagram below shows an isosceles triangle. Find the missing angles in this isosceles triangle, labelled x.



[2 marks]

Sum of angles in a triangle = 180°. Base angles of isosceles triangle are equal:

$2x = 180° - 78° = 102°$ [M1]

$x = 102° \div 2 = 51°$ [A1]

Answer $x = 51°$ ✓

14 Jessie measures the heights of her 7 friends, and works out that their mean height is 1.73 m. She measures her own height and works out that the mean height of all of them is 1.74 m.

What is Jessie's height?

[3 marks]

Step 1: Calculate total height of the 7 friends:

Total height (7 friends) = 7 × 1.73 m = 12.11 m [M1]

Step 2: Calculate total height of all 8 people (including Jessie):

Total height (8 people) = 8 × 1.74 m = 13.92 m [M1]

Jessie's height = 13.92 m − 12.11 m = 1.81 m [A1]

Answer 1.81 m ✓

Turn over for next question

5

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15 Demi pre-booked a hire car to use on holiday in Spain. She paid £70 in total which gave her car hire for 7 days.

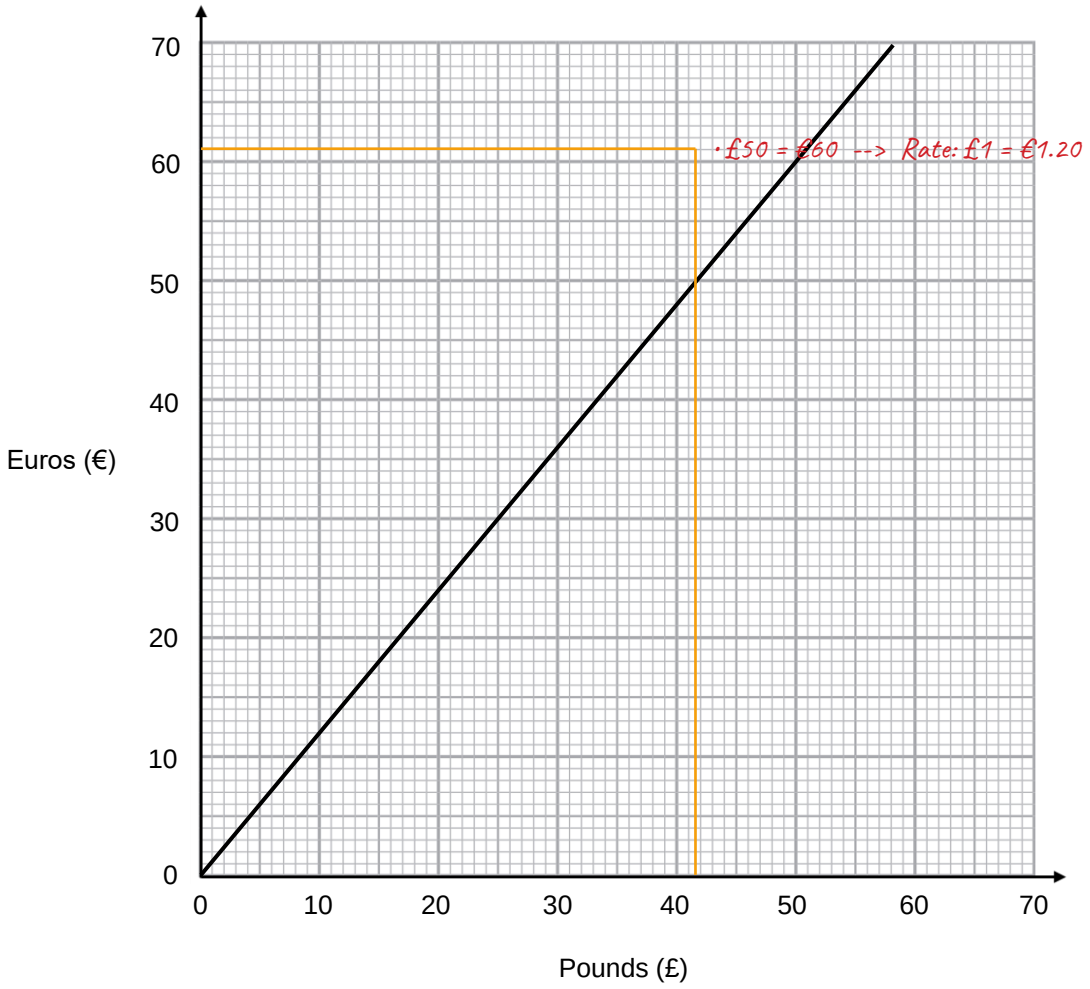
When she went on holiday, she realised that it may have been cheaper to hire the car in Spain, instead of pre-booking it.

The cost of the car hire, in Euros (€), in Spain is calculated using this formula:

$$\text{cost} = 0.6(30 + 15d)$$

where d is the number of days the car is hired for.

Demi uses this conversion graph to convert between Pounds (£) and Euros (€).



Work how much Demi would have saved by paying for the car hire in Spain, instead of pre-booking it. Give your answer in Pounds (£).

[5 marks]

Answer space provided on next page

5

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15 cont

Step 1: Calculate car hire cost in Spain (€) using formula (d = 7 days):

Cost (€) = 0.6 × (30 + 15 × 7) = 0.6 × (30 + 105) = 0.6 × 135 = €81.00 [M1 M1]

Step 2: Read exchange rate from conversion graph on previous page:

From graph: £50 = €60 --> Exchange rate = 60 ÷ 50 = €1.20 per £1 [M1]

Step 3: Convert €81 to Pounds (£):

Cost in £ = €81 ÷ 1.20 = £67.50 [M1]

Step 4: Calculate savings compared to pre-booking (£70.00):

Savings = £70.00 - £67.50 = £2.50 [A1]

Answer £2.50 ✓

16 Whilst Demi is on holiday in Spain, she buys a coat that was reduced by 20% in a sale. If the sale price was €180, how much was its original price?

Give your answer in Euros (€).

[2 marks]

Reverse percentage: Sale price represents 100% - 20% = 80% (0.80):

Original price = €180 ÷ 0.80 = €225.00 [M1 A1]

Answer €225 ✓

Turn over for next question

2

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17 The table gives information about the number of girls and boys in Years 9 and 10.

	Year 9	Year 10	Total
Boys	62	97	159
Girls	112	84	196
Total	174	181	355

[3 marks for table]

Complete the two-way table and use this to find the probability that a student selected at random is a boy.

[4 marks]

Find probability that a randomly chosen student is a boy:

Total Boys = 159 | Grand Total Students = 355

P(Boy) = 159 / 355 [M1 A1]

Answer 159 / 355 ✓

18 Susan makes fudge using 2 parts of chocolate to 3 parts of frosting. Susan makes a batch of fudge using 510 g of frosting.

She eats 10% of the fudge she has made, and then splits the remaining fudge into individual bars, each weighing 100 g.

How many individual bars of fudge can she make?

[3 marks]

Step 1: Calculate total mass of fudge (2 parts chocolate : 3 parts frosting):

1 part = 510 g ÷ 3 = 170 g --> Total (5 parts) = 5 × 170 g = 850 g [M1]

Step 2: Remaining fudge after eating 10% (90% remaining):

Remaining fudge = 850 g × 0.90 = 765 g [M1]

Step 3: Number of 100 g individual bars = 765 g ÷ 100 g = 7.65 --> 7 bars [A1]

Answer 7 individual bars ✓

Turn over for next question

7

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19 Donald is a zookeeper. In 2018, he measured the weights of the 6 leopards that are kept in the zoo, and recorded his results in the table below.

Leopard	1	2	3	4	5	6
Weight (kg)	32	29	27	34	30	27

In 2021, the median weight of the leopards was 68 lbs.

1 kg = 2.20 lbs

Calculate the percentage change in the median weight of the leopards from 2018 to 2021.

[5 marks]

Step 1: Find 2018 median weight (order data values in kg):

Ordered values: 27, 27, [29, 30], 32, 34

Median (2018) = (29 + 30) ÷ 2 = 29.5 kg **[M1 A1]**

Step 2: Convert 2018 median weight to lbs (1 kg = 2.20 lbs):

29.5 kg × 2.20 = 64.9 lbs **[M1]**

Step 3: Calculate percentage change from 2018 (64.9 lbs) to 2021 (68 lbs):

Percentage Change = [(68 − 64.9) ÷ 64.9] × 100%

= [3.1 ÷ 64.9] × 100% = 4.7765...% ≈ 4.78% increase **[M1 A1]**

Answer **4.78% increase** ✓

Turn over for next question

5

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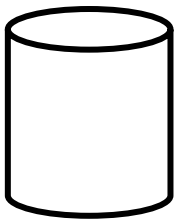
20 A paperweight is shaped as a cylinder. The paperweight has a radius of 2 cm and a height of 5 cm.



The paperweight has a mass of 350 g. Calculate its density in g/cm^3.

Use $\pi = 3.14$.

[4 marks]

Step 1: Calculate volume of cylinder: $V = \pi r^2 h$ (radius = 2 cm, height = 5 cm):

$V = 3.14 \times 2^2 \times 5 = 3.14 \times 4 \times 5 = 3.14 \times 20 = 62.8\ cm^3$ **[M1 A1]**

Step 2: ~~Calculate density: Density = Mass ÷ Volume:~~

Density = 350 g ÷ 62.8 cm^3 = 5.5732... g/cm^3 **[M1]**

Rounding to 2 decimal places = 5.57 g/cm^3 **[A1]**

Answer **5.57 g/cm^3** ✓

Turn over for next question

4

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21 Nabeela travelled to London by train. Her journey was made up of two parts, including a short wait in between.

The journey consisted of a train to York and then another train to London from York.

The first train departed at 07:05 and the journey lasted for 12 minutes.

The second train departed at 07:27 and the journey lasted for 122 minutes.

The total distance travelled was 197 miles.

Including the time waiting between trains, what was Nabeela's average speed for the entire journey?

Give your answer in miles per hour (mph). **[4 marks]**

Step 1: Calculate total journey time including waiting time:

- First train (07:05 departure): lasts 12 mins (arrives 07:17)
- Wait at York: 07:17 to 07:27 = 10 mins
- Second train (07:27 departure): lasts 122 mins

Total Time = 12 + 10 + 122 = 144 minutes [M1]

Step 2: Convert total time from minutes to hours:

Time in hours = 144 ÷ 60 = 2.4 hours [M1]

Step 3: Calculate average speed: Speed = Distance ÷ Time:

Average Speed = 197 miles ÷ 2.4 hours = 82.0833... mph ≈ 82.08 mph [M1 A1]

Answer 82.08 mph ✓

Turn over for next question

4

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22 A shop has three offers on cola.

Offer 1	Offer 2	Offer 3
2 litre bottle	4 × 330 ml cans	500 ml bottle
Price: £1.80 Offer: 10% off	Price: £2.20 Offer: half price	Price: 60p Offer: $\frac{1}{4}$ off

Which offer provides the best value for money?

Show your working.

[4 marks]

Calculate unit cost per litre for each promotional offer:

- **Offer 1 (2 L @ £1.80, 10% off):**
  Price = £1.80 × 0.90 = £1.62 --> Unit cost = £1.62 ÷ 2 = £0.81 per litre **[M1]**
- **Offer 2 (4 × 330 ml = 1.32 L @ £2.20, half price):**
  Price = £2.20 × 0.50 = £1.10 --> Unit cost = £1.10 ÷ 1.32 = £0.833... per litre **[M1]**
- **Offer 3 (500 ml = 0.5 L @ 60p, 1/4 off):**
  Price = 60p × 0.75 = 45p (£0.45) --> Unit cost = £0.45 ÷ 0.5 = £0.90 per litre **[M1]**

Comparison: £0.81 < £0.833... < £0.90 --> Offer 1 gives the lowest cost per litre. **[A1]**

Answer **Offer 1 provides the best value for money (£0.81/L)** ✓

End of Section B

4

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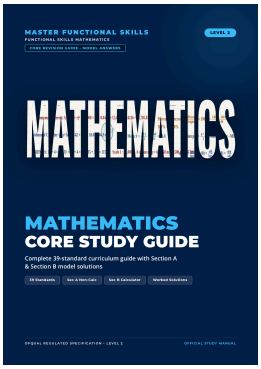

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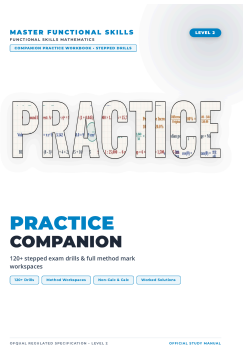

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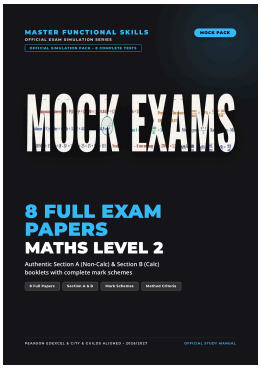

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